



## ISOLATION OF NATURAL DYE FROM CODIAEUM VARIEGATUM

**Khalida Tasneem<sup>\*</sup> , Rajiv Khatioda**

Department of Chemistry,  
North Eastern Regional Institute of Science and Technology  
(Deemed University), Itanagar , India

### Abstract

The powdered dried leaves of *Codiaeum Variegatum* “ garden croton” or Variegated croton” refluxed with alcohol for 12 hours. Natural dye obtained was then applied on wool using different mordants gave beautiful parrot green, light green, light brown and dark brown shades.

*Codiaeum Variegatum* is a species of plant in the genus *Codiaeum*, which is a member Of the family Euphorbiaceae. It is native to southern India, Sri Lanka, Indonesia, Malaysia and the Western Pacific Ocean islands, growing in open forests and scrub. It is an evergreen shrub growing to 3m tall and has large, thick, leathery , shiny evergreen leaves, alternately arranged, 30 cm long and 0.5-8 cm broad.

Preparation of Natural Dye and its application on wool with different mordants :-

The macerated leaves of *Codiaeum Variegatum* was refluxed with alcohol for 12 hours ,filtered and the solvent was evaporated in rotator apparatus under reduced pressure, the solid residue obtained as natural dye was treated with wool with 10% and 20% of mordants copper sulphate, Ferrous sulphate and potassium dichromate. Boiling wool with natural dye and mordant for one hour. Washing of dyed wool with water and than drying in shade at room temperature. In all cases same timing and same procedure was applied. Mordanting helps the dye bind to the fibre. Different mordants allow different colour ranges for each dyestuff.

The amount of mordant used was a percentage of the weight of fibre.It was important to note the different in shades when the wool was first mordanting then dyeing, than the colour of wool first dyeing then mordanting.Also the shades changes with the increase in the percentage of modants taking the percentage of dye same. There was also change in shades when first dyeing and than mordanting.

Key words: Codiaeum Variegatum, dye,

## OBSERVATION AND SHADE CARD

### 5.1 Observations:

| Preparation of mordants |                   |                   |                      |             |
|-------------------------|-------------------|-------------------|----------------------|-------------|
| Mordants                | 10% of 200mg wool | 20% of 200mg wool | Amount of water (ml) | Observation |
| Copper sulphate         | 20mg              | 40mg              | 100                  | Blue        |
| Ferrous sulphate        | 20mg              | 40mg              | 100                  | Yellow      |
| Potassium dichromate    | 20mg              | 40mg              | 100                  | Orange      |

| Mordants with dye         |                      |                      |                       |                       |                           |                           |
|---------------------------|----------------------|----------------------|-----------------------|-----------------------|---------------------------|---------------------------|
| Dye                       | 20mg copper sulphate | 40mg copper sulphate | 20mg ferrous sulphate | 40mg ferrous sulphate | 20mg potassium dichromate | 40mg potassium dichromate |
| 20 mg Codiaeum Variegatum | Light yellow colour  | Parrot green colour  | Muddy colour          | Muddy colour          | Yellow colour             | Yellow colour             |

**Colour of wool 1<sup>st</sup> mordanting then dyeing :**

| Sl. No | Sample                                                                            | Composition ( Mordant + Dye )      |
|--------|-----------------------------------------------------------------------------------|------------------------------------|
| 1      |  | 10% Copper Sulphate + 10% dye      |
| 2      |  | 20% Copper Sulphate + 10% dye      |
| 3      |  | 10% Ferrous Sulphate + 10% dye     |
| 4      |  | 20% Ferrous Sulphate + 10% dye     |
| 5      |  | 10% Potassium dichromate + 10% dye |
| 6      |  | 20% Potassium dichromate + 10% dye |

**Colour of wool 1<sup>st</sup> Dyeing then mordanting :**

| Sl. No | Sample                                                                              | Composition (Dye+ Mordant)         |
|--------|-------------------------------------------------------------------------------------|------------------------------------|
| 1      |  | 10% dye + 10% Copper Sulphate      |
| 2      |  | 10% dye + 20% Copper Sulphate      |
| 3      |  | 10% dye + 10% Ferrous Sulphate     |
| 4      |  | 10% dye + 20% Ferrous Sulphate     |
| 5      |  | 10% dye + 10% Potassium dichromate |
| 6      |  | 10% dye + 20% Potassium dichromate |

## **CONCLUSION :-**

The dye- bath colour was not representative of the ending wool colour. With the higher concentration of mordant the wool seemed to absorb the dye better , thus dyeing darker. The variation of mordants created a variety of different effects in wool colour wise. The mordants didn't effect the texture of the wool. Mordants have been proved to be useful chemicals in the natural dyeing process.

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